

Model Cards and Human Review

Project: Responsible Plant Classifier

1. Your mission

Your learning target

I can write a model card whose claims match evidence.

I can separate intended use from unsupported use.

I can review a card for missing evaluation details.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Proxy Features and Fairness. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Overfitting. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

A model card defines intended use, evidence, limitations, risks, decision boundaries, monitoring, and human escalation.



LEARNING TARGET Draft five model-card headings for a classroom leaf classifier. Why must the card state that its labels are classroom categories rather than professional plant diagnosis?

Builder name and experiment date:

Readiness for the connected explanation

What should remain visible after an evaluation?

This week's end goal

By the end, I can explain model cards and human review using a traced example and a stated limitation.

2. Notice the evidence

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible evidence; do not explain more than the picture and information support.



MYSTERY The team must use model cards and human review to solve a model-building problem. What information do they have? What important evidence is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what evidence would confirm or reject your prediction.

My prediction and supporting evidence:

A closer explanation

A model card helps another person understand what was built and what evidence exists. It is a record, not a certificate of safety. Link statements to saved data and run records. Use unknown or not evaluated where evidence is absent instead of filling gaps with optimistic language.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines model cards and review.

3. See how it works

ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.



IN

INPUT

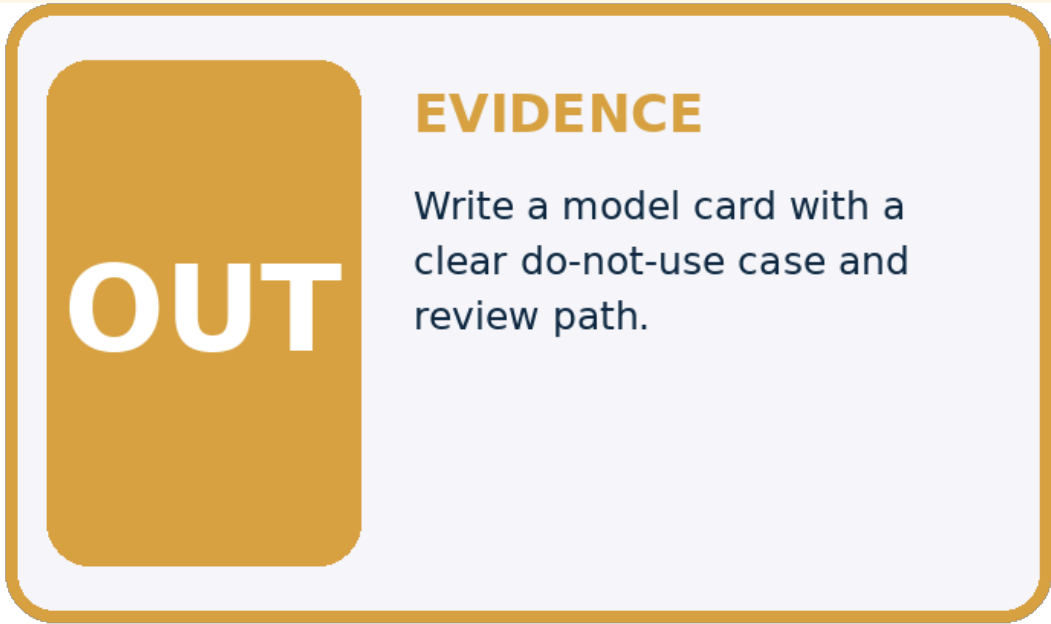
Write a model card with a clear do-not-use case and review path.



ML

MODEL

A model card defines intended use, evidence, limitations, risks, decision boundaries, monitoring, and human escalation.



OUT

EVIDENCE

Write a model card with a clear do-not-use case and review path.

QUESTION Write a model card with a clear do-not-use case and review path.

MODEL IDEA A model card defines intended use, evidence, limitations, risks, decision boundaries, monitoring, and human escalation.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Intended use: classroom exploration of visible leaf spotting. Evidence: test metrics, confusion matrix, subgroup slices, and data version.

Draw a fresh data → model → prediction → evaluation trace:

A closer explanation

Example card: classroom parcel classifier v 1; purpose: compare toy classification methods on fictional clear/blocked labels. Training 40, validation 20, test 12 with separate record groups. Frozen test TN 7 FP 1 FN 2 TP 2, blocked positive: accuracy 9/12, precision 2/3, recall 2/4. Always-clear baseline scores 8/12. No real parcel or student decision is supported.

Before continuing, underline the decision rule and circle the evidence that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim. Keep the distinction between an observed result and a proposed next step.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL CASE

Expected data from the intended setting



BOUNDARY CASE

Value near a threshold or class boundary



MISSING CASE

Unknown field, label, or necessary evidence



SHIFTED CASE

New subgroup, device, condition, or time

NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain prediction.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change: The publicity sentence becomes “80% accurate for every plant”.

Prediction: Reject that unbounded statement.

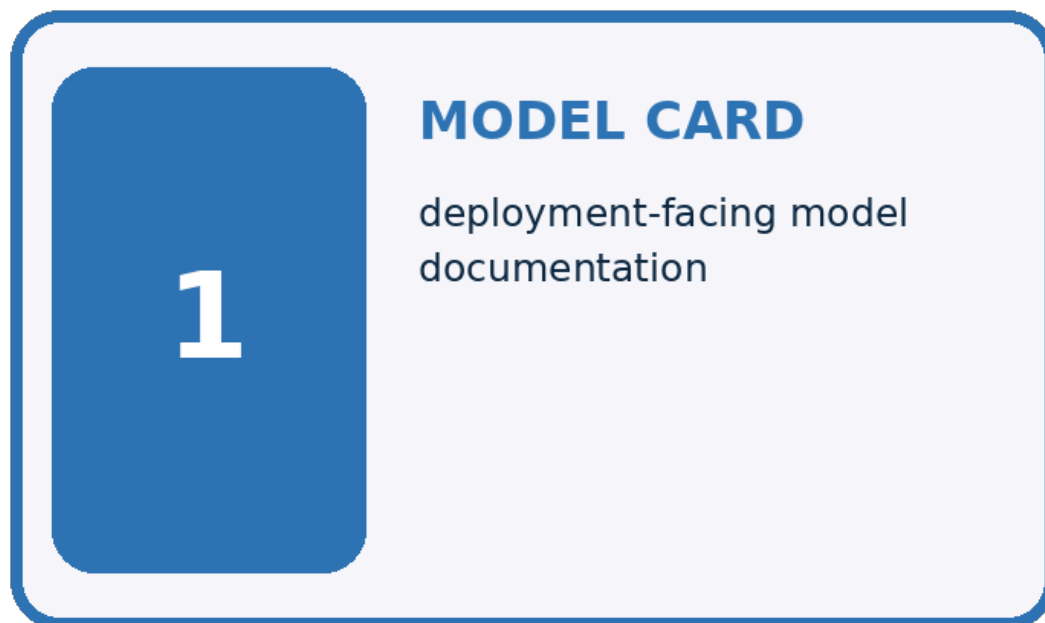
Reason: The measured rate belongs to twenty related indoor photos, not every plant or setting.

5. Words that unlock the idea

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.



2

INTENDED USE

approved purpose and setting

3

DO-NOT-USE

explicit prohibited application

4

HUMAN REVIEW

authorized person checks or decides

Term · Precise meaning · My ML example

Term: model card

Precise meaning: deployment-facing model documentation

My ML example: _____

Term: intended use

Precise meaning: approved purpose and setting

My ML example: _____

Term: do-not-use

Precise meaning: explicit prohibited application

My ML example: _____

Term: human review

Precise meaning: authorized person checks or decides

My ML example: _____

MISCONCEPTION TO REPAIR A disclaimer at the bottom of the screen is a complete safety plan.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word · Meaning

Word: Model card

Meaning: A concise structured account of a model and its limits

Word: Intended use

Meaning: The purpose and conditions the evidence supports

Word: Out-of-scope use

Meaning: A purpose for which the model is not supported

Word: Review status

Meaning: The current state of checks and unresolved issues

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

WEEK 33 • WORKED EVIDENCE

- 1 Intended use: classroom exploration of visible leaf spotting.
- 2 Evidence: test metrics, confusion matrix, subgroup slices, and data version.
- 3 Do not use for pesticide, medical, or agricultural diagnosis.
- 4 Uncertain, unfamiliar, or high-consequence cases route to a teacher or qualified adult.

STEP 1 Intended use: classroom exploration of visible leaf spotting.

STEP 2 Evidence: test metrics, confusion matrix, subgroup slices, and data version.

STEP 3 Do not use for pesticide, medical, or agricultural diagnosis.

STEP 4 Uncertain, unfamiliar, or high-consequence cases route to a teacher or qualified adult.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

A closer explanation

Model cards and review at a glance

Part · Evidence or action · Interpretation

Part: Purpose and version

Evidence or action: Toy classifier v 1

Interpretation: Fictional data only

Part: Evidence

Evidence or action: 12 held-out cases

Interpretation: Full matrix and baseline

Part: Limits and review

Evidence or action: No real decisions

Interpretation: List unresolved coverage

The model card should name feature availability, preprocessing fitted on training, model settings, reference-label rules and test date. Include subgroup coverage if evaluated; otherwise say it was not assessed. Describe relevant failure examples without exposing personal information.

Verify the worked accuracy, precision and recall. Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Model Cards and Human Review

Write a model card whose performance statement can be traced to a concrete evaluation.

The facts we will use

Toy $v = 3$ classifier scored 16/20 on five unseen plants with four views each, under indoor tablet lighting. Baseline scored 14/20.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Model Cards and Human Review

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name evidence

Record $v = 3$, 20 photo cases, 5 plant identities, group-separated split and indoor tablet conditions.



2. Compare baseline

$v = 3$ accuracy 80%; baseline 70% on the same 20 cases.



3. State limits

Do not claim all cameras, outdoor conditions or disease diagnosis were validated.

CHANGE ONE THING

The publicity sentence becomes "80% accurate for every plant".

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Name evidence

What happens and why: Record $v = 3$, 20 photo cases, 5 plant identities, group-separated split and indoor tablet conditions.

Step: 2. Compare baseline

What happens and why: $v = 3$ accuracy 80%; baseline 70% on the same 20 cases.

Step: 3. State limits

What happens and why: Do not claim all cameras, outdoor conditions or disease diagnosis were validated.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Do not claim all cameras, outdoor conditions or disease diagnosis were validated. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

A model card communicates evidence; it is not certification of safety, fairness or suitability for consequential plant decisions.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



COMPARE • DIAGNOSE • REPAIR
Tempting claim: A disclaimer at the bottom of the screen is a complete safety plan. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: passes

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: defined behavior

Actual: _____

Likely cause / next test: _____

Evidence record: missing case

Expected: safe fallback

Actual: _____

Likely cause / next test: _____

Evidence record: shifted case

Expected: measured limit

Actual: _____

Likely cause / next test: _____

CLAIM TO CHALLENGE A disclaimer at the bottom of the screen is a complete safety plan.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Do not claim a synthetic classroom dataset represents every school or age group. Missing calibration, subgroup or shift evidence must remain explicit. A useful card may conclude that the model should stay a classroom demonstration.

What should change in a card after a threshold revision?

8. Practise together

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



FREEZE THE EXPERIMENT

Record question, intended use, data version, grouped split, preprocessing, features, model settings, seed, metric, predictions, errors, subgroup results, and human review before claiming improvement.

Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/preprocessing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

A peer reviewer traces one headline number back to the saved predictions, checks denominators and looks for mismatches between intended use and evidence. Record requested changes and whether each was resolved. A card must be updated when its model, data or intended use changes.

Practise with less help: Check arithmetic

Use workbook W2 for the connected example “Model cards and review”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Model cards and review. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Write a bounded replacement for the publicity sentence. Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Model Card Clinic: compare two cards, find unsupported claims and missing gates, then author a release-ready version.

1. Write the hypothesis and expected evidence before running.
2. Model Card Clinic: compare two cards, find unsupported claims and missing gates, then author a release-ready version.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

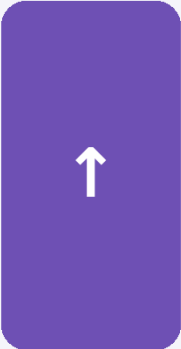
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the example.



ML BUILDER MASTERY
1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

DEFINE Use all four terms precisely: model card, intended use, do-not-use, human review.

CALCULATE Reproduce the worked evidence with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE A disclaimer at the bottom of the screen is a complete safety plan.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed evidence.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

11. Connect your project

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same evidence-based capstone.

**RESPONSIBLE PLANT CLASSIFIER**

Complete the Plant Classifier model card and require teacher sign-off before any classroom demonstration.

THIS WEEK'S CHECKPOINT Complete the Plant Classifier model card and require teacher sign-off before any classroom demonstration.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My evidence-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



INDEPENDENT ML BUILDER PROOF

Write a model card with a clear do-not-use case and review path.

REQUIRED PROOF Write a model card with a clear do-not-use case and review path.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent evidence and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

NEXT BRIDGE Carry the experiment evidence into: Capstone Dataset and Baseline.

13. Study a complete case

Week 33 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A model card says healthy/spotted,80% accuracy only.

The question

Add responsible-use evidence.

Worked explanation

Intended users/conditions, labels/data, evaluation sets/metrics, class/subgroup results, versions, limitations, excluded uses and human review. Classroom classification is not professional diagnosis.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

A model card supports responsible review

Evidence or stage · Value or result

Evidence or stage: Purpose

Value or result: Users, conditions and excluded uses

Evidence or stage: Evidence

Value or result: Data, labels, metrics and subgroup results

Evidence or stage: Version

Value or result: System and evaluation identity

Evidence or stage: Limits

Value or result: Failure conditions and human review

A model card says healthy/spotted, 80% accuracy only.

Add responsible-use evidence.

Intended users/conditions, labels/data, evaluation sets/metrics, class/subgroup results, versions, limitations, excluded uses and human review. Classroom classification is not professional diagnosis.

14. Try a new case independently

Independent case: Apply independently

Review this card: model v 2, 90 percent accurate, works for every school. No dataset size, split or baseline is given. Write a correction request with a bounded replacement claim.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 33 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A model card says 90% accuracy but gives no intended use, data conditions or human-review rule. What should be added before classroom use?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Complete the Plant Classifier model card and require teacher sign-off before any classroom demonstration.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Draft five model-card headings for a classroom leaf classifier. Why must the card state that its labels are classroom categories rather than professional plant diagnosis?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.